Title: Robot programming and simulation for machining (cutting ,welding)

Objective: To do robot programming and simulation for deburring

Procedure:

1. Launch RoboDK and create a new project.

2. Select File➔ Open online library (Ctrl+Shift+O). A new nested window will appear showing the online library, It is also possible to select the corresponding button in the toolbar.



3.Use the filters to find your robot by brand, payload,

4.Select Download. The robot should automatically appear in the station in a few seconds.

5.The online library can be closed once the robot is loaded.

6 First we will import Table 1400x800x800mm, UR16e Base, ATI 9150-CDB-8-11-000 Deburring Tool (Spring), and random1 object

7. Select Program➔ Add Reference Frame

Alternatively, select the equivalent button in the toolbar.

Double click the reference frame (on the tree or on the 3D geometry on the main screen) to enter the coordinates shown in the image (X,Y,Z position and Euler angles for the orientation). The mouse wheel can be used on top of each case to quickly update the position of the reference frame on the main screen.

The following colors are used by default:

- X coordinate ➔ Red
- Y coordinate ➔ Green
- Z coordinate ➔ Blue
- 1st Euler rotation ➔ Cyan
- 2nd Euler rotation ➔ Magenta
- 3rd Euler rotation ➔ Yellow

8. Adjust the robot model's position and orientation within the virtual environment to match the desired setup.

9. Gripper and Tool Definition:

a. Define the robot's gripper within RoboDK by specifying its dimensions, grip functionality, and attachment point on the robot arm.

b. Configure the Tool Center Point (TCP) for the robot to ensure accurate positioning of the gripper during box manipulation.

10.Create targets:

Follow these steps to create two targets as a new home target and approach target respectively:

a.Double click the robot to show the robot panel

b.Select gripper as the Tool Frame. Once a tool or a reference frame becomes active it will show a green dot in the tree icon.

c.Select Frame 2 as the Reference Frame

d.Hold the Alt key and move the robot by dragging it through the TCP or the robot flange to a safe position, free of collisions with any objects. Alternatively, move the coordinates of the Tool Frame (TCP) with respect to the reference Frame.

e.Use the Other configurations section to switch between different robot configurations and make sure that none of the robot axes are close to the axis limits

11. Select **Program➔** **Teach Target** (Ctrl+T), or the corresponding button in the toolbar (as shown in the image). The target will be placed as a dependency of the active reference frame and will automatically remember the current robot position (cartesian and joints axes).

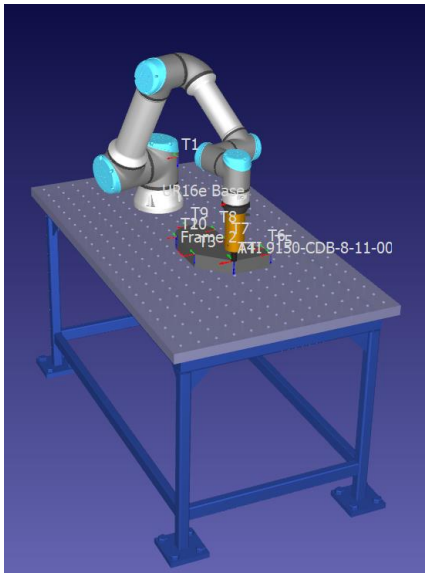
12. You can rename the target by pressing **F2**. Alternatively, select **Tools➔Rename item**.

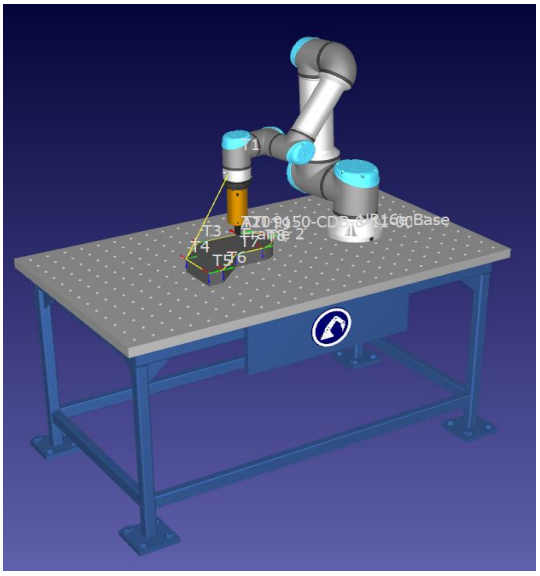
13. Add a program:-

a. Select **Program➔** **Add Program**, or the appropriate button in the toolbar.

b. create a program using **Program➔** **Move Joint Instruction** (or the corresponding button in the toolbar) , **Program➔** **Move Linear Instruction**, or the appropriate button in the toolbar , Attach and Detach .

16. Save the project then run and simulate it.





Conclusion: RoboDK software provides a comprehensive platform for programming and simulating industrial robots for deburring tasks. By leveraging its capabilities, such as modeling, tool definition, programming interface, and simulation environment, we can develop a practical solution for robot programming and simulation in deburring applications. The software-based approach enables thorough testing and optimization of the deburring program, leading to improved efficiency and quality in the real-world implementation.